

Notes: Glosten and Milgrom (JFE, 1985)

Bid, Ask and Transaction Prices in a Specialist Market with Heterogeneously Informed Traders

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model primitives and timing	4
3.1	Trading protocol	4
3.2	Security value and information	4
3.3	Trader preferences and reservation value	5
4	Zero-profit bid and ask quotes	5
4.1	Specialist expected profit	5
4.2	Competitive zero-profit condition	5
5	Proposition 1: the bid and ask straddle expected value	6
5.1	Statement	6
5.2	Derivation	6
6	Proposition 2: transaction prices form a martingale	6
6.1	Transaction price as revised expectation	6
6.2	Martingale proof	6
6.3	Implication for serial correlation	7
7	Adding noninformational trading costs	7
8	Proposition 3: volume times squared spread is bounded	7
8.1	Statement	7
8.2	Derivation intuition	8
9	Proposition 4: specialist and trader expectations converge	8
9.1	Statement	8
9.2	Derivation intuition	8
9.3	Inventory implication	9
10	Proposition 5: comparative statics of the spread	9
10.1	Statement	9
10.2	Explanation	9
11	Example: two values and perfect insider information	9
11.1	Environment	9
11.2	Inelastic liquidity demand	10
11.3	Uniform liquidity parameter	10
11.4	Learning speed	10

12 Market breakdown	10
13 Discounting and the distinction between observed and realizable returns	11
13.1 Renormalized reservation value	11
13.2 Expected realizable return condition	11
14 Empirical implications and tests	11
15 Original equation and proposition panels	12
15.1 Investor order rule and specialist zero-profit conditions	12
15.2 Proposition 1 and Proposition 2	12
15.3 Serial correlation and spread-volume bound	13
15.4 Learning and inventory balance	14
15.5 Comparative statics and two-state example	14
15.6 Bayesian learning speed and realizable returns	15
16 Short summary	16
17 Conclusion	17
17.1 Core conclusion	17
17.2 Economic intuition	17
17.3 Incremental contribution revisited	17
17.4 Implications for empirical work	17
17.5 Limitations	17
17.6 Takeaway	17

1 Big picture

- **Question.** Why can a positive bid-ask spread exist even when the market maker is risk neutral, faces no explicit order-processing or inventory costs, and earns zero expected profits on each side of the market?
- **Core answer.** A spread arises because of adverse selection. A buy order is more likely when the arriving trader has favorable private information; a sell order is more likely when the arriving trader has unfavorable private information. The specialist therefore sets the ask equal to the expected value conditional on a buy and the bid equal to the expected value conditional on a sell.
- **Market setting.** The model is a pure dealership/specialist market. The specialist posts a bid and an ask for a unit trade. A trader arrives anonymously, observes the quotes, and buys one unit, sells one unit, or does not trade.
- **Information structure.** Some traders are informed about the eventual value V of the security. Others are liquidity traders whose willingness to buy or sell comes from a private preference/liquidity parameter. The specialist observes public history but not the arriving trader's type or private signal.
- **Pricing rule.** In competitive zero-profit equilibrium,

$$A_t = \mathbb{E}[V \mid S_t, Z_t > A_t], \quad B_t = \mathbb{E}[V \mid S_t, Z_t < B_t].$$

The bid and ask are not arbitrary transaction-cost markups; they are posterior expectations conditional on the trade direction.

- **Main dynamic result.** Transaction prices form a martingale. With a purely adverse-selection spread, bid-to-ask bouncing does not mechanically produce negative serial correlation in transaction price changes.
- **Information aggregation.** Trading itself reveals information. As enough trades occur, the specialist's expectation and informed traders' conditional expectations converge.
- **Spread-volume relation.** The expected value of volume times the average squared spread is bounded by the variance of the underlying value. This implies that high-volume markets should have smaller adverse-selection spreads.
- **Market breakdown.** If informed traders are too numerous, too well informed, or liquidity demand/supply is too elastic, the bid and ask can widen so much that trade shuts down.
- **Return measurement implication.** Observed returns based on transaction prices can exceed realizable returns available to uninformed traders. The wedge is the expected loss of uninformed traders to informed traders.

2 Incremental contribution of the paper

- **A canonical adverse-selection foundation for the bid-ask spread.** Earlier market-maker models emphasized inventory costs, order-processing costs, or dealer monopoly power. Glosten and Milgrom show that private information alone is enough to generate a spread.

- **Quote prices as conditional expectations.** The paper gives a precise Bayesian interpretation of the bid and ask: the ask is the expected value conditional on a buy order, and the bid is the expected value conditional on a sell order.
- **Dynamic learning through trades.** The model is not only a static lemons story. It describes how each transaction updates beliefs and how transaction prices process private information over time.
- **Martingale transaction prices despite a spread.** This is a subtle and important distinction. A spread created by adverse selection is not observationally equivalent to a spread created by order-processing costs or dealer rents.
- **A way to interpret serial correlation.** When both adverse-selection and noninformational trading-cost components exist, the serial correlation of transaction price changes depends on the fraction of the spread due to noninformational costs.
- **Volume-spread bound.** The paper gives a formal upper bound on expected volume times squared average spread that does not depend on the detailed pattern of insider activity.
- **Market failure under asymmetric information.** The model shows that a competitive zero-profit specialist market may shut down, echoing the Akerlof lemons logic in a dynamic securities setting.
- **Return-measurement contribution.** The paper links microstructure spreads to the wedge between observed returns and realizable returns, with implications for small-firm and neglected-firm return anomalies.

3 Model primitives and timing

3.1 Trading protocol

- A specialist posts a bid B_t and an ask A_t for one unit of the security.
- One trader arrives anonymously.
- The trader observes the quote and either buys one unit at A_t , sells one unit at B_t , or does not trade.
- After the trader acts, the trade direction and transaction price become part of the public/specialist history. The specialist can revise quotes before the next arrival.

Interpretation. This is a sequential market. A quote is a menu conditional on the next arriving order. A buy order and a sell order reveal different information, so the specialist rationally attaches different posterior values to the ask and bid.

3.2 Security value and information

- The eventual value per share is a random variable $V \geq 0$, with finite variance.
- At a future date T_0 , enough public information is revealed that informational differences disappear.
- Public information before a trader's arrival is denoted H_t .

- The specialist's information is denoted S_t .
- If the arriving trader is informed, he also observes a private signal J_t .
- The trader's information after seeing the quote is denoted F_t .

3.3 Trader preferences and reservation value

All agents are risk neutral. A trader has utility

$$\rho_t x V + c, \quad (1)$$

where x is the number of shares held, c is current consumption, and ρ_t is a trader-specific liquidity/preference parameter. The specialist is normalized to have $\rho = 1$.

Given information F_t , the trader's reservation value for one share is

$$Z_t = \rho_t \mathbb{E}[V \mid F_t]. \quad (2)$$

The order decision is

$$\text{buy if } Z_t > A_t, \quad \text{sell if } Z_t < B_t, \quad (3)$$

with no trade otherwise.

Economic meaning of ρ_t . The liquidity parameter makes trade possible despite common rationality. If everyone had identical preferences and all trade were purely informational, no-trade logic would be too strong. Heterogeneous ρ_t generates genuine liquidity demand and supply.

4 Zero-profit bid and ask quotes

4.1 Specialist expected profit

For a proposed ask A and bid B , the specialist's expected profit from the next arrival is

$$\mathbb{E} \left[(A - V) \mathbf{1}_{\{Z_t > A\}} + (V - B) \mathbf{1}_{\{Z_t < B\}} \mid S_t \right]. \quad (4)$$

Equivalently,

$$\begin{aligned} & (A - \mathbb{E}[V \mid S_t, Z_t > A]) \mathbb{P}(Z_t > A \mid S_t) \\ & - (B - \mathbb{E}[V \mid S_t, Z_t < B]) \mathbb{P}(Z_t < B \mid S_t). \end{aligned} \quad (5)$$

4.2 Competitive zero-profit condition

The paper imposes the key zero-profit equilibrium conditions

$$A_t = \mathbb{E}[V \mid S_t, Z_t > A_t], \quad (6)$$

$$B_t = \mathbb{E}[V \mid S_t, Z_t < B_t]. \quad (7)$$

That is, the specialist breaks even separately on the ask side and on the bid side.

Core intuition. When a trader buys at the ask, the specialist learns that the trader's reservation value is high. Since informed traders buy when their signal is favorable, a buy order raises the conditional expected value of the security. The specialist must therefore set the ask above the pre-trade expected value. The logic is symmetric for sell orders and the bid.

5 Proposition 1: the bid and ask straddle expected value

5.1 Statement

If the zero-profit bid and ask exist, then

$$A_t \geq \mathbb{E}_t[V] \geq B_t. \quad (8)$$

The inequalities are strict if adverse selection is possible.

5.2 Derivation

Let C be the event that a customer buys:

$$C = \{Z > A\} = \{\mathbb{E}[V | F] > A/\rho\}. \quad (9)$$

By the zero-profit ask definition,

$$A = \mathbb{E}[V | S, C]. \quad (10)$$

Taking expectations conditional on C and using iterated expectations,

$$A = \mathbb{E}[A | C] = \mathbb{E}[\mathbb{E}[V | S, C] | C] = \mathbb{E}[V | C] \quad (11)$$

$$= \mathbb{E}[\mathbb{E}[V | C, \rho] | C] = \mathbb{E}[\mathbb{E}[\mathbb{E}[V | F, C, \rho] | C, \rho] | C] \quad (12)$$

$$= \mathbb{E}[\mathbb{E}[\mathbb{E}[V | F] | C, \rho] | C]. \quad (13)$$

Because C occurs when $\rho\mathbb{E}[V | F] > A$, conditioning on C selects high posterior values of V . Therefore,

$$A = \mathbb{E}[V | C] \geq \mathbb{E}[V]. \quad (14)$$

A symmetric argument yields $B \leq \mathbb{E}[V]$.

Economic message. The spread is the price of learning from order flow. The ask is high because a buy order is good news about V . The bid is low because a sell order is bad news about V .

6 Proposition 2: transaction prices form a martingale

6.1 Transaction price as revised expectation

Let T_k be the time of the k th transaction and let p_k be the transaction price. If the k th trade is at the ask, the transaction price is A_{T_k} ; if it is at the bid, it is B_{T_k} . Because the bid and ask are conditional expectations,

$$p_k = \mathbb{E}[V | S_k], \quad (15)$$

where S_k denotes the specialist's information immediately after the k th trade.

6.2 Martingale proof

For the specialist's information filtration $\{S_k\}$,

$$\mathbb{E}[p_{k+1} | S_k] = \mathbb{E}[\mathbb{E}[V | S_{k+1}] | S_k] \quad (16)$$

$$= \mathbb{E}[V | S_k] = p_k. \quad (17)$$

Thus transaction prices form a martingale. Since public information H_k is contained in specialist information and p_k is publicly observed, the same martingale property holds relative to public transaction histories.

6.3 Implication for serial correlation

A purely adverse-selection spread does not by itself imply negative serial correlation in transaction price changes. This differs from order-processing-cost models, where prices bounce between bid and ask around an unchanging value and mechanically induce negative autocorrelation.

Economic message. In the adverse-selection model, a trade at the ask is not simply a mechanically high price. It also means that beliefs about V have moved up. That information effect offsets bid-ask bounce.

7 Adding noninformational trading costs

Suppose each trade costs the specialist c . Then zero profit becomes

$$A = \mathbb{E}[V \mid S_t, Z_t > A] + c, \quad (18)$$

$$B = \mathbb{E}[V \mid S_t, Z_t < B] - c. \quad (19)$$

Let Ψ be the adverse-selection component of the spread,

$$\Psi = \mathbb{E}[V \mid S_t, Z_t > A] - \mathbb{E}[V \mid S_t, Z_t < B]. \quad (20)$$

The total spread is $\Psi + 2c$. Let

$$\beta = \frac{2c}{\Psi + 2c} \quad (21)$$

be the fraction of the spread due to noninformational trading costs. With suitable symmetry assumptions, the correlation coefficient of adjacent transaction price changes is

$$R = -\frac{\beta}{\delta + \beta^2}, \quad (22)$$

where

$$\delta = 1 + \left(\frac{2\theta}{\Psi + 2c} \right)^2 \quad (23)$$

and θ^2 is the variance of public information arriving between trades.

Economic message. Negative serial correlation identifies the noninformational component of the spread. If the spread is mostly adverse selection, the price process learns from trade and serial correlation is closer to zero.

8 Proposition 3: volume times squared spread is bounded

8.1 Statement

Define

$$\frac{1}{\gamma_k} = \mathbb{P}(Z_k > A_k \mid S_{T_k}) \mathbb{P}(Z_k < B_k \mid S_{T_k}). \quad (24)$$

Let γ^* be the average of γ_k over the $N + 1$ relevant observations. Define the average spread over N trades as

$$\Psi_N = \frac{1}{N} \sum_{k=1}^N (A_k - B_k). \quad (25)$$

Then

$$\mathbb{E} \left[\frac{N}{N+1} \frac{\Psi_N^2 N}{\gamma^*} \right] \leq \text{Var}(V). \quad (26)$$

If $\gamma^* \leq \gamma$ almost surely, then

$$\mathbb{E}[N\Psi_N^2] \leq 2 \text{Var}(V)\gamma. \quad (27)$$

8.2 Derivation intuition

Since $p_{N+1} = \mathbb{E}[V \mid S_{N+1}]$, the variance of V bounds the variance of the posterior transaction-price martingale:

$$\text{Var}(V) \geq \text{Var}(p_{N+1}). \quad (28)$$

Because $\{p_k\}$ is a martingale, its increments are orthogonal:

$$\text{Var}(p_{N+1}) = \mathbb{E} \left[\sum_{k=1}^{N+1} (p_k - p_{k-1})^2 \right]. \quad (29)$$

The conditional variance of the next price move is tied to the spread. If the probabilities of a buy and sell are not too close to zero, a wide spread implies a large posterior jump. Summing these jumps over many trades cannot generate more total variance than the variance of V . This produces the bound.

Economic message. Private information is finite. Spreads are large when trades are very informative, but once many informative trades occur, the market learns and spreads must shrink.

9 Proposition 4: specialist and trader expectations converge

9.1 Statement

If trade is reasonably balanced, then

$$\mathbb{E}[V \mid S_k] - \mathbb{E}[V \mid F_k] \rightarrow 0 \quad (30)$$

in probability as the number of trades grows.

9.2 Derivation intuition

Let

$$D = \mathbb{E}[V \mid F] - \mathbb{E}[V]. \quad (31)$$

When a trader buys, the ask satisfies

$$A - \mathbb{E}[V] \geq \mathbb{E}[D \mid D > \mathbb{E}[V](1 - \rho)/\rho]. \quad (32)$$

If the informed trader's posterior could remain substantially above the specialist's posterior with positive probability, a buy order would continue to move the ask by a positive amount. But Proposition 3 says the spread and posterior jumps must vanish with enough balanced trade. Therefore the specialist's posterior and the trader's posterior converge.

9.3 Inventory implication

As consensus emerges, expected supply and expected demand balance:

$$\lim \{\mathbb{P}(Z_k < B_k \mid H_k) - \mathbb{P}(Z_k > A_k \mid H_k)\} = 0. \quad (33)$$

Thus the specialist's inventory tends toward a driftless process.

Economic message. Trades are information events. The specialist need not observe the insider's signal directly; by observing enough buy and sell decisions, he asymptotically learns what insiders know.

10 Proposition 5: comparative statics of the spread

10.1 Statement

Holding the trading history fixed, the ask price rises and the bid price falls when:

1. insider information becomes finer;
2. the ratio of informed to uninformed arrivals increases;
3. the elasticity of uninformed demand and supply increases.

10.2 Explanation

Better insider information makes order direction more informative, increasing adverse selection. More informed arrivals relative to liquidity arrivals also increases adverse selection. In both cases, the specialist must protect himself by raising the ask and lowering the bid.

The elasticity result is subtler. With more elastic liquidity supply and demand, liquidity traders become more responsive to price concessions. To make enough money from liquidity traders to offset informed-trader losses, the specialist must widen the spread in the model's stated direction for quote bounds.

Dynamic qualification. The proposition is local in time and history. If insider arrival rates are higher, the immediate spread rises. But higher insider trading also accelerates information revelation, which can reduce future spreads.

11 Example: two values and perfect insider information

11.1 Environment

The example assumes

$$V \in \{1, 11\}, \quad \mathbb{P}(V = 11) = \pi, \quad (34)$$

so the expected value is

$$E = 11\pi + 1(1 - \pi). \quad (35)$$

Insiders know V perfectly and have $\rho = 1$. The fraction of arrivals that are insiders is α .

11.2 Inelastic liquidity demand

Liquidity traders have $\rho = 0$ or $\rho = \infty$ with probability one-half each. The ask solves

$$\alpha\pi(11 - A) = 0.5(1 - \alpha)(A - E). \quad (36)$$

The left side is the specialist's expected loss to informed buyers. The right side is expected profit from liquidity buyers.

The bid solves

$$\alpha(1 - \pi)(B - 1) = 0.5(1 - \alpha)(E - B). \quad (37)$$

11.3 Uniform liquidity parameter

If $\rho \sim U(0, 2)$, the ask solves

$$\alpha\pi(11 - A) = (1 - \alpha) \left[1 - \frac{A}{2E} \right] (A - E), \quad (38)$$

provided $A \leq 2E$. The bid solves

$$\alpha(1 - \pi)(B - 1) = (1 - \alpha) \left[\frac{B}{2E} \right] (E - B). \quad (39)$$

11.4 Learning speed

Bayes' rule updates the odds ratio:

$$\frac{\pi^+}{1 - \pi^+} = \frac{\pi}{1 - \pi} \times \text{Factor}, \quad (40)$$

where the likelihood ratio Factor depends on whether the arriving trader buys or sells.

In the inelastic case, the expected number of trades needed until nearly all information is revealed is approximately

$$\frac{\log(\text{Odds}) + (1 - \pi_0) \log(\pi_0 / (1 - \pi_0))}{\alpha \log((1 + \alpha) / (1 - \alpha))}. \quad (41)$$

For small α , the spread is approximately proportional to α , while the time to reveal information is approximately proportional to $1/\alpha^2$.

Economic message. More insider activity widens the current spread but speeds up learning. This is why average spread and information revelation cannot be understood from one trade alone.

12 Market breakdown

If informed traders are too numerous or too well informed relative to liquidity trading, the specialist may be unable to quote any ask at which he can break even. In the uniform-liquidity example, an ask may fail to exist because the quadratic equation for A has no acceptable root.

Economic message. Competitive zero-profit intermediation may fail under severe adverse selection. A high spread discourages liquidity trades; the absence of liquidity trades prevents information from being revealed; the lack of information keeps the spread high. The market can remain closed until information arrives from outside the trading process.

13 Discounting and the distinction between observed and realizable returns

13.1 Renormalized reservation value

The paper later introduces a discounting normalization:

$$Z_t^* = \exp(r_t(T_0 - t))Z_t. \quad (42)$$

The zero-profit condition becomes

$$A_t^* = \exp[-r_t(T_0 - t)]\mathbb{E}[V \mid S_t^*, Z_t^* > A_t^*], \quad (43)$$

$$B_t^* = \exp[-r_t(T_0 - t)]\mathbb{E}[V \mid S_t^*, Z_t^* < B_t^*]. \quad (44)$$

13.2 Expected realizable return condition

Let τ be a holding period and suppose the expected realizable return of an uninformed trader over τ is i :

$$\mathbb{E}_t \left[\frac{B_{t+\tau}^*}{A_t^*} \right] = e^{i\tau}. \quad (45)$$

Proposition 6 shows that the discount rate can be written

$$r_t = i + \frac{n+1}{T_0 - t} \log(k_t), \quad (46)$$

where

$$\frac{1}{k_t} = \left\{ \mathbb{E}_t \left[\frac{B_{t+\tau}}{A_t} \frac{B_{t+2\tau}}{A_{t+\tau}} \dots \frac{B_{t+(n+1)\tau}}{A_{t+n\tau}} \right] \right\}^{1/(n+1)} \leq 1. \quad (47)$$

Since $1/k_t$ is the expected geometric mean gross return from repeatedly buying at asks and selling at bids in the no-discounting market, it is less than or equal to one. Thus the premium term measures the expected loss of uninformed traders to informed traders.

Observed versus realizable returns. Returns computed from transaction prices can exceed returns that a real uninformed trader can earn. A real buyer pays the ask and later sells at the bid. Observed prices may be mid-like realized transaction prices that do not represent executable round-trip returns.

14 Empirical implications and tests

- **Spread determinants.** Spreads should be larger when private information is more important, insider arrivals are more likely, and liquidity trading is less able to absorb adverse selection.
- **Spread-volume relation.** Average adverse-selection spreads should be lower in high-volume markets because more trades reveal information.
- **Serial correlation decomposition.** Negative serial correlation in transaction price changes indicates noninformational spread components such as order-processing costs or dealer rents. A purely adverse-selection spread produces a martingale transaction-price sequence.

- **Market closure.** In thin markets or periods of unusually severe asymmetric information, quoted spreads may widen enough to suppress trade.
- **Return measurement.** Small-firm and neglected-firm return anomalies may partly reflect the difference between observed transaction returns and realizable returns after bid-ask costs and adverse selection.
- **Information events.** Around earnings announcements, takeover rumors, or other events that temporarily advantage insiders, spreads should widen and then shrink as trades reveal information or public announcements resolve uncertainty.

15 Original equation and proposition panels

The paper has no standalone empirical tables or conventional figures. The panels below therefore extract the main original equation and proposition blocks from the paper. They are arranged two per row, following the same visual layout convention as previous notes.

15.1 Investor order rule and specialist zero-profit conditions

Read:

- Panel (a) shows the investor's order rule: buy if $Z_t > A$, sell if $Z_t < B$, where $Z_t = \rho_t \mathbb{E}[V | F_t]$.
- Panel (b) shows the specialist's expected-profit expression and the resulting zero-profit bid and ask restrictions.

Economic message: The core of the model is the meeting of two conditional expectations: the trader conditions on private information and liquidity needs, while the specialist conditions on what the trader's action reveals.

expected profit from an arrival at time t is

$$\mathbb{E}[(A - V)I_{\{Z_t > A\}} + (V - B)I_{\{Z_t < B\}} | S_t], \quad (2)$$

where $I_{\{Z_t > A\}}$ and $I_{\{Z_t < B\}}$ are, respectively, the indicator functions of the events $\{Z_t > A\}$ and $\{Z_t < B\}$, i.e., $I_{\{Z_t > A\}}$ is one if the event $\{Z_t > A\}$ occurs; otherwise it is zero. The expression, (2), may be rewritten as

$$(A - \mathbb{E}[V | S_t, Z_t > A])\mathbb{P}\{Z_t > A | S_t\} - (B - \mathbb{E}[V | S_t, Z_t < B])\mathbb{P}\{Z_t < B | S_t\}, \quad (3)$$

where $\mathbb{E}[\cdot | \cdot]$ is the (conditional) expectation operator derived from the probability measure $\mathbb{P}$.

The above holds as long as there are zero costs associated with all short positions in cash or stock. Our central assumption about the specialist is that he earns zero expected profits on each purchase and each sale, and he faces no transaction costs. To illustrate how competition might lead to such a description, suppose there are two specialists in this one stock. Both have the same information and face the same population. Suppose the first specialist sets an ask price A^1 so that $A^1 > \mathbb{E}[V | S_t, Z_t > A^1]$. The second specialist will rationally undercut the first by choosing an ask $A^2 < A^1$ and $A^2 \geq \mathbb{E}[V | S_t, Z_t > A^2]$. The zero expected profit equilibrium at time t (if it exists) consists of a pair of functions A_t and B_t satisfying

$$\begin{aligned} A_t(\omega) &= \mathbb{E}[V | S_t, Z_t > A_t(\omega)](\omega), \\ B_t(\omega) &= \mathbb{E}[V | S_t, Z_t < B_t(\omega)](\omega), \end{aligned} \quad (4)$$

Putting the utility functions and information structures together, the optimal decision of an investor arriving at time t , given bid B and ask A is given by

buy if $Z_t > A$,
sell if $Z_t < B$,

where Z_t is given by

$$Z_t = \rho_t \mathbb{E}[V | F_t] = \rho_t(1 - U_t)\mathbb{E}[V | H_t, J_t, A, B] + \rho_t U_t \mathbb{E}[V | H_t, A, B], \quad (1)$$

where U_t is one if the individual arriving at t is uninformed and zero otherwise.

(a) Investor decision rule and Z_t

(b) Specialist profit and zero-profit quotes

15.2 Proposition 1 and Proposition 2

Read:

- Panel (a) states and proves the spread-straddling result: $A_t \geq E_t[V] \geq B_t$.
- Panel (b) states the martingale result for transaction prices.

Economic message: A buy order is good news and a sell order is bad news. Because transaction prices themselves are revised posterior expectations, they form a martingale.

expected profit conditions:

$$A_t = E[V|S_t, Z_t > A_t],$$

$$B_t = E[V|S_t, Z_t < B_t].$$

Then the ask price is greater and the bid price is less than the expectation of V : $A_t \geq E_t[V] \geq B_t$. The inequalities are strict if adverse selection is possible, i.e., if

$$P\{Z_t > E_t[V], E_t[V|F_t] > E_t[V]\} > 0,$$

$$P\{Z_t < E_t[V], E_t[V|F_t] < E_t[V]\} > 0.$$

Proof. We prove only the first inequality, since the proof of the second is similar. Also, for brevity, we omit the time subscripts. Let C be the event that the customer makes a purchase, i.e., the event that Z is greater than A .

$$C = \{Z > A\} = \{E[V|F] > A/\rho\}.$$

Then, by definition, $A = E[V|S, C]$ so

$$\begin{aligned} A &= E[A|C] = E[E[V|S, C]|C] = E[V|C] = E[E[V|C, \rho]|C] \\ &= E[E[E[V|F, C, \rho]|C, \rho]|C] = E[E[E[V|F, \rho]|C, \rho]|C] \\ &= E[E[E[V|F]|C, \rho]|C] \geq E[E[E[V|F]|\rho]|C] \\ &= E[E[V]|\rho] = E[V] \end{aligned}$$

(a) Proposition 1: bid and ask straddle value

takes place at the ask at time t , i.e., there is an arrival at time t and Z_t exceeds A_t , then the transaction price will be the ask price, which in this event is equal to the revised expectation of V given this event. Similarly, if there is a trade at the bid, the transaction price is the bid price which in this case is also the revised expectation of V given this (different) event. Mathematically, the transaction price is given by $A_k I_{\{Z_k > A_k\}} + B_k I_{\{Z_k < B_k\}}$. This is, by definition, equal to

$$E[V|S_{T_k}, Z_{T_k} > A_{T_k}] I_{\{Z_{T_k} > A_{T_k}\}} + E[V|S_{T_k}, Z_{T_k} < B_{T_k}] I_{\{Z_{T_k} < B_{T_k}\}}. \quad (5)$$

But (5) is just $E[V|S_{T_k}^+] = E[V|S_k]$. This observation allows us to write the k th transaction price as $p_k = E[V|S_k]$, as long as k trades take place (i.e., as long as T_k is less than T_0). Thus, if N trades actually take place, then $p_1, \dots, p_N$ are the prices at which trades occur. Notice, however, that p_{N+1} is also well defined, and represents some intermediate value between the bid and ask prices at the end of the trading period. If one thinks in terms of computing returns based on daily data, this use of an imaginary transactions price is not far from the way CRSP calculations are done for days in which there is no trade. The specification of H_t^+ from the preceding paragraph implies that p_k is measurable with respect to H_k which allows us to prove the following proposition:

Proposition 2. The sequence of transaction prices $\{p_k\}$ forms a martingale relative to the specialist's information, $\{S_k\}$, and the public information, $\{H_k\}$.

(b) Proposition 2: transaction-price martingale

15.3 Serial correlation and spread-volume bound

Read:

- Panel (a) displays the serial-correlation expression when the spread combines adverse selection and transaction costs.
- Panel (b) gives Proposition 3, bounding expected volume times squared average spread.

Economic message: Serial correlation and spread-volume behavior help distinguish adverse-selection spreads from order-processing or dealer-rent spreads.

The variance of the price change is given by the following: $\theta^2 + (\Psi/2)^2 + c\Psi + 2c^2$, where θ^2 is the variance of public information arriving exogenously between trades.⁶ Let β be the proportion of the spread due to transaction costs; i.e., $\beta = 2c/(\Psi + 2c)$. Then the correlation coefficient, R , is given by the following:

$$R = -\beta/(\delta + \beta^2), \quad (6)$$

where $\delta = 1 + (2\theta/(\Psi + 2c))^2$. The correlation coefficient thus moves toward zero if the proportion of the spread due to informational asymmetries increases. We can invert (6) to get the proportional of the spread due to trading costs as a function of the correlation coefficient and δ (a measure of how much public information arrives relative to the spread),

$$\beta = (-1 + (1 - 4\delta R^2)^{1/2})/2R. \quad (7)$$

Unfortunately, (7) alone does not define a useful statistic since in general δ will not be known. It does suggest the possibility that transaction data might be used to find a measure of informational asymmetry.

(a) Serial correlation formulas

Every time there is a transaction, record the bid and ask that prevailed just prior to the trade. Do this for a unit of time, and denote the number of trades observed by N . The following proposition shows that if trade is reasonably balanced (i.e., the probability of a purchase given that a trade occurred is bounded away from zero and one) then the expectation of the number of trades times the average spread squared is bounded by a number that is independent of the pattern of trade.⁷

Proposition 3. Define

$$1/\gamma_k = P\{Z_k > A_k | S_{T_k}\} P\{Z_k < B_k | S_{T_k}\},$$

and let γ^* be the mean value of γ_j over the $N+1$ observations; i.e., $\gamma^* = \sum_{j=1}^{N+1} \gamma_j / (N+1)$ (if $N+1$ trades do not occur prior to T_0 , put $\gamma_{N+1} = 4$). Further, define Ψ_N to be the average spread over N trades, i.e.,

$$\Psi_N = \sum_{k=1}^N (A_k - B_k) / N.$$

Then,

$$E[(N/(N+1)) \Psi_N^2 N / \gamma^*] \leq \text{var}(V).$$

In particular, if there is a number γ such that $P\{\gamma^* < \gamma\} = 1$, then

$$E[(N/(N+1)) N \Psi_N^2] \leq \text{var}(V) \gamma,$$

and hence, if N is almost surely positive, the expected value of the volume times average spread squared is bounded by a number that is independent of the pattern of trade,

$$E[N \Psi_N^2] \leq 2 \text{var}(V) \gamma.$$

(b) Proposition 3: volume times squared spread

15.4 Learning and inventory balance

Read:

- Panel (a) gives Proposition 4: specialist and trader expectations converge as balanced trade accumulates.
- Panel (b) gives the inventory corollary: expected buy and sell probabilities balance in the limit.

Economic message: The specialist learns from trades. Once enough order-flow information is incorporated, the specialist no longer faces systematic inventory drift from asymmetric information.

of that tendency toward consensus.

Proposition 4. If trade is reasonably balanced in the sense of Proposition 3, i.e., the probability of a purchase is bounded away from zero and one, then the expectations of the specialist and the traders converge as the number of trades increases, i.e., $E[V|S_k] - E[V|F_k]$ converges to zero in probability (where F_k is the information of the k th trader to arrive).

Proof. We use the notation and results of Proposition 1 and two general facts from probability theory cited earlier. The suppressed time subscript is now $t = T_k$.

$$\begin{aligned} A &= E[V|C] = E[E[V|F, \rho]|C] = E[E[V|F]]E[V|F] > A/\rho \\ &> E[E[V|F]]E[V|F] > E[V]/\rho. \end{aligned}$$

Define D by

$$D = E[V|F] - E[V].$$

(a) Proposition 4: convergence of expectations

using bid prices shows that $P\{D_t \leq -\epsilon\}$ goes to zero. Thus, $E_k[V|F_k] - E_k[V]$ converges in probability to zero. Also, $E_k[V|S_k] = E_k[V]$ (both are equal to A_k if the customer buys and B_k if he sells). Q.E.D.

As the consensus described in Proposition 4 emerges, there comes to be a balance between expected supply and expected demand:

Corollary 1. The specialist's inventory of stocks tends to a driftless stochastic process; i.e.,

$$\lim P\{Z_k < B_k | H_k\} - P\{Z_k > A_k | H_k\} = 0.$$

Proof. Using the results of the Proposition 4, let $\hat{p}$ be the limit of $E[V|S_k]$ and let $\hat{Z} = \rho \hat{p}$. Then,

$$\begin{aligned} \lim (P\{Z_k < B_k | H_k\} - P\{Z_k > A_k | H_k\}) &= P\{\hat{Z} < \hat{p} | \hat{S}\} - P\{\hat{Z} > \hat{p} | \hat{S}\} \\ &= P\{\alpha < 1\} - P\{\alpha > 1\} = 0 \end{aligned}$$

(b) Corollary: inventory becomes driftless

15.5 Comparative statics and two-state example

Read:

- Panel (a) gives Proposition 5: better insider information, more informed arrivals, and supply/demand elasticity affect quotes.
- Panel (b) shows the example's break-even equations for ask and bid prices under inelastic and uniform liquidity-trader preferences.

Economic message: The spread is larger when order flow is more informative about hidden value. The example translates the abstract Bayesian logic into explicit equations.

Proposition 5 provides proof of results also reported in Copeland and Galil (1983, p. 1463).

Proposition 5. For any given time t , the ask price A_t increases and the bid price B_t decreases when, other things being equal,

- (i) the insiders' information at time t becomes better (i.e., finer),
- (ii) the ratio of informed to uninformed arrival rates at t is increased, or
- (iii) the elasticity of uninformed supply and demand at time t increases.

Proof. If there is an arrival at time t , we can express the reservation price Z_t by

$$Z_t = (1 - U_t)E[V|H_t, J_t] + U_t \rho_t E[V|H_t],$$

where [as in (1) above] U_t is one if an arrival at t is uninformed and zero otherwise, and J_t is the information of an insider if an insider arrives at t (for the succeeding discussion, time subscripts have been dropped to simplify the notation). Define M by $M = E[V|J]$ where J is the information set of an arriving insider, and let G be the distribution function of the liquidity parameter (and let g be its density). Denote demand and supply elasticities of the uninformed by e_D and e_S . Then e_D and e_S are given by

$$e_D = Ag(A/E[V])/E[V](1 - G(A/E[V])),$$

$$e_S = Bg(B/E[V])/E[V]G(B/E[V]).$$

Recalling [from (4a)] that A and B , the ask and bid, are the smallest a and largest b satisfying

$$a \geq E[V|Z > a], \quad b \leq E[V|Z < b],$$

then, for ask price A and bid price B ,

$$AP\{Z > A\} - E[V|_{Z > A}] \geq 0,$$

(a) Proposition 5: spread comparative statics

at that price, then the specialist will not sell A to the market. The event just described will occur with probability $\alpha\pi$, since the event that the buyer is an insider is independent of the value of V . The probability that a liquidity trader will buy at an ask price of A is precisely the probability that the preference parameter ρ exceeds A/E , which in the inelastic case is 0.5 and in the uniform case is $1 - A/(2E)$. So the specialist's break-even condition in the inelastic case is the linear equation (in A),

$$\alpha\pi(11 - A) = 0.5(1 - \alpha)(A - E), \quad (8)$$

and in the uniform case is the quadratic equation,

$$\alpha\pi(11 - A) = (1 - \alpha)[1 - A/(2E)](A - E), \quad (9)$$

provided $A \leq 2E$. The smaller root of the quadratic equation is the relevant one here, since the ask price is the *lowest* price at which the specialist breaks even.

A similar argument shows that the bid price for the inelastic case solves the linear equation

$$\alpha(1 - \pi)(B - 1) = 0.5(1 - \alpha)(E - B), \quad (10)$$

while the bid price for the uniform case is the larger root of the quadratic equation

$$\alpha(1 - \pi)(B - 1) = (1 - \alpha)[B/(2E)](E - B). \quad (11)$$

(b) Example: bid and ask break-even equations

15.6 Bayesian learning speed and realizable returns

Read:

- Panel (a) shows the Bayes odds update and the approximate number of trades needed to reveal nearly all information.
- Panel (b) gives Proposition 6's discounting result linking the normal return to the loss of uninformed traders to informed traders.

Economic message: Information is revealed through a sequence of trade signs. Observed transaction returns can include a compensation component that uninformed traders cannot realize on a round trip.

specialist after a trade has just occurred. These posterior beliefs also serve as the prior beliefs for determining the bid and ask prices for the next trader. π^+ is determined from Bayes' Theorem using the formula

$$[\pi^+/(1-\pi^+)] = [\pi/(1-\pi)] \text{ Factor},$$

where *Factor* is the likelihood ratio whose numerator is the probability of the trader's action given that $V=1$ and whose denominator is the probability of the action given $V=1$. Note that *Factor* may depend on the action taken, the bid price, the ask price, π and α , as well as on the model used – uniform or inelastic.

For any given model and any value of the parameter α , the endogenous variable π determines the expected value E , the bid and ask prices B and A , the spread, the probability that the next trader will buy or sell, etc. Then, from the formula for π^+ above, it is clear that the stochastic process of values of π over time is a Markov process; given the current value of π , its future distribution is independent of its history. Moreover, since π_k is a conditional probability given all information up to time T_k , the Markov process is a martingale. In addition, in the inelastic case, $\log(\pi_k(1-\pi_0)/\pi_0(1-\pi_k))/\log(1+\alpha)/(1-\alpha)$ is, given V , a random walk whose value is the accumulated excess of purchases over sales by arriving customers.

Let us say that “nearly all of the insider information has been assimilated in the prices” when we reach the point where $\pi/(1-\pi) < 1/Odds$ when V is low or $\pi/(1-\pi) > Odds$ when V is high. For the case of perfectly inelastic demand, one can show⁸ that the expected number of trades that must take place before nearly all information is revealed is approximately⁹

$$\frac{\log(Odds) + (1-\pi_0)\log(\pi_0/(1-\pi_0))}{\alpha \log((1+\alpha)/(1-\alpha))}. \quad (12)$$

For small values of α , (12) is approximately proportioned to $1/\alpha^2$. Also for small values of α and for any π , the size of the spread determined in (8) and (10) is approximately proportional to α . Thus, the effect of doubling α from, say, 0.1 to 0.2 is roughly to double the spread at each level of π and to divide

(a) Bayes updating and learning speed

~ ~ ~ ~ ~ with uniformly distributed values of π and α and with heterogeneous expectations

$B_{t+\tau}^* = Ve^{-i(T_0-t-\tau)}$ (i.e., $r_{t+\tau} = i$ for $t \in [T_0-\tau, T_0]$). Then r_t , the discount rate at t , is the normal return i plus a premium,

$$r_t = i + (n+1)/(T_0-t)\log(k_t),$$

where

$$\frac{1}{k_t} = \left\{ E_t \left[\frac{B_{t+\tau}}{A_t} \frac{B_{t+2\tau}}{A_{t+\tau}} \dots \frac{B_{t+(n+1)\tau}}{A_{t+n\tau}} \right] \right\}^{1/(n+1)} \leq 1,$$

and

$$t + n\tau \in [T_0 - \tau, T_0].$$

The discount rate applied at time t , r_t , has a particularly interesting interpretation. Notice that $(1/k_t)$ is the expected geometric mean gross return per τ units of time earned by an investor that follows a strategy of buying and selling every τ periods of time in a market with no discounting. The log of this is thus the continuously compounded expected return from such a strategy. Obviously, such a return is negative. Recall from the definition of A_t and B_t that the specialist sets the bid and ask so that on average what he loses to the informed is made up by what he gains from the uninformed liquidity traders. Thus, $(n+1)/(T_0-t)\log(k_t)$ (a positive number) is, in return (per unit time) terms, what the uninformed on average lose to the informed. Thus, r_t represents the expected holding period return i plus the return that the uninformed

(b) Proposition 6: normal return plus adverse-selection premium

16 Short summary

- A positive bid-ask spread can arise purely from asymmetric information.
- The specialist sets the ask equal to expected value conditional on a buy, and the bid equal to expected value conditional on a sell.
- Therefore the ask exceeds, and the bid is below, the pre-trade expected value.
- Transaction prices form a martingale because each transaction price is a posterior expectation of V .
- A spread caused only by adverse selection does not mechanically imply negative serial correlation in transaction price changes.
- Adding order-processing costs or dealer profits creates negative serial correlation; the correlation contains information about the noninformational component of the spread.
- The expected value of volume times squared average spread is bounded by the variance of fundamental value.
- As trade accumulates, the market learns insider information, so specialist and trader expectations converge.
- If adverse selection is too severe, the market can shut down because no break-even quote induces trade.
- Observed transaction returns can exceed realizable returns because uninformed traders buy at asks and sell at bids while informed traders earn the adverse-selection premium.

17 Conclusion

17.1 Core conclusion

Glosten and Milgrom provide the classic sequential-trade explanation for the bid-ask spread. The spread is a rational response to adverse selection: order direction reveals information, and the specialist must price that information into quotes.

17.2 Economic intuition

A buy order is not just a demand shock. It is a signal that the trader may know the asset is worth more. A sell order is a signal that the trader may know the asset is worth less. A competitive specialist therefore sets different break-even prices for buying and selling.

17.3 Incremental contribution revisited

The paper's central innovation is to connect microstructure, Bayesian learning, and asset-return measurement. It shows that quotes, trades, serial correlation, volume, information aggregation, and apparent return anomalies are all linked through the same adverse-selection mechanism.

17.4 Implications for empirical work

Empirical returns calculated from transaction prices can be misleading, especially for thinly traded or informationally opaque stocks. Realizable returns must account for the fact that an uninformed trader pays the ask when buying and receives the bid when selling.

17.5 Limitations

The model abstracts from limit orders, strategic dynamic order splitting, specialist risk aversion, explicit inventory costs, order-processing costs, and endogenous information acquisition. These omissions make the model clean but also mean that actual spreads usually combine several components.

17.6 Takeaway

The Glosten-Milgrom model is the foundation for adverse-selection microstructure theory: spreads are compensation for trading against better-informed agents, and transaction prices are the mechanism through which private information becomes public.